

1. Using the initial condition,

$$\frac{1}{4} = -\frac{1}{4} + \frac{1}{4} + Ce^{-1},$$

so

$$C = \frac{e}{4}.$$

Therefore, the particular solution is

$$y(x) = -\frac{1}{4} + \frac{x}{2} + \frac{1}{4}e^{1-2x}.$$

2. Differentiate the proposed solution:

$$y' = Ce^x + 4 \sin(2x) - 4 \cos(2x).$$

On the other hand,

$$y + 5 \sin(2x) = Ce^x - 2 \cos(2x) + 3 \sin(2x).$$

These expressions are not equal in general. Therefore,

No, it is not the general solution.

3. For

$$y' = f(x, y) = x - 4y, \quad y(0) = -\frac{3}{4},$$

we have $x_0 = 0$, $y_0 = -3/4$, and $h = 0.2$.

- (a) Euler's method:

$$y_{i+1} = y_i + hf(x_i, y_i).$$

Thus,

$$y_1 = -\frac{3}{4} + 0.2 \left(0 - 4 \left(-\frac{3}{4} \right) \right) = -0.15,$$

$$y_2 = -0.15 + 0.2(0.2 - 4(-0.15)) = 0.01,$$

$$y_3 = 0.01 + 0.2(0.4 - 4(0.01)) = 0.082.$$

Therefore,

$$(x_0, y_0) = (0, -0.75), \quad (x_1, y_1) = (0.2, -0.15), \quad (x_2, y_2) = (0.4, 0.01), \quad (x_3, y_3) = (0.6, 0.082).$$

- (b) Mark the four points

$$(0, -0.75), \quad (0.2, -0.15), \quad (0.4, 0.01), \quad (0.6, 0.082)$$

on the graph and connect consecutive points by straight-line segments.

- (c) For the midpoint method,

$$y_{i+1} = y_i + h f \left(x_i + \frac{h}{2}, y_i + \frac{h}{2} f(x_i, y_i) \right),$$

$$x_{i+1} = x_i + h.$$

For the first step,

$$f(0, -0.75) = 3,$$

so at the midpoint,

$$f(0.1, -0.75 + 0.1(3)) = f(0.1, -0.45) = 1.9.$$

Hence

$$y_1 = -0.75 + 0.2(1.9) = -0.37.$$

For the second step,

$$f(0.2, -0.37) = 1.68,$$

and at the midpoint,

$$f(0.3, -0.37 + 0.1(1.68)) = f(0.3, -0.202) = 1.108.$$

Thus

$$y_2 = -0.37 + 0.2(1.108) = -0.1484.$$

Therefore,

$$(x_0, y_0) = (0, -0.75), \quad (x_1, y_1) = (0.2, -0.37), \quad (x_2, y_2) = (0.4, -0.1484).$$

(d) At $x_2 = 0.4$, the exact value is

$$y(0.4) \approx -0.101304.$$

For Euler's method,

$$E_E = |0.01 - (-0.101304)| = \boxed{0.111304}.$$

For the midpoint method,

$$E_M = |-0.1484 - (-0.101304)| = \boxed{0.047096}.$$

4. For

$$f(x, y) = xy + \frac{1}{x}, \quad h = \frac{1}{12},$$

the four Runge-Kutta values are

$$k_1 = x_i y_i + \frac{1}{x_i},$$

$$k_2 = \left(x_i + \frac{1}{24}\right) \left(y_i + \frac{1}{24}k_1\right) + \frac{1}{x_i + \frac{1}{24}},$$

$$k_3 = \left(x_i + \frac{1}{24}\right) \left(y_i + \frac{1}{24}k_2\right) + \frac{1}{x_i + \frac{1}{24}},$$

$$k_4 = \left(x_i + \frac{1}{12}\right) \left(y_i + \frac{1}{12}k_3\right) + \frac{1}{x_i + \frac{1}{12}}.$$